

Components of Custom Charge Detection Mass Spectrometer

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Project: Custom Computer Controlled Radiofrequency (RF) Generator for Quadrupole Ion Guides

Description: Design and construct custom RF generators capable of generating 1-1.5 kV amplitude waveforms in the frequency range of DC to 10 MHz that can drive capacitive loads up to 1 nF. The design will consist of three components: A computer interface that can synthesize arbitrary waveforms, a power amplifier for synthesized waveforms and a center-tapped step-up transformer for the power amplified waveform. The computer interface will communicate with either Windows or Linux operating systems through a serial peripheral interface and will require custom software drivers for controlling the RF generator from the connected computer. A design for the power amplifier exists online however it will need to be constructed, tested and potentially modified for the application. The step-up transformer will be center-tapped to generate two voltage amplified waveforms that are 180 degrees out of phase and will need to be wound with a 1:20 ratio. The RF generator will be tested first on the bench with an oscilloscope and appropriate load and then with a custom-built charge detection mass spectrometer here at Berkeley. A total of three units is required for instrument control.

Project: Custom Parallelized Miniature Linear Array Detector for High-throughput Charge Detection Mass Spectrometer

Design and construct a prototype custom 3-dimensional parallelized miniature array detector for high-throughput (10,000+ ions per second) mass analysis of individual ions. The custom array detector will consist of ~24 metal (conductive) detector elements each of which are ~2.0 mm thick with a diameter of ~5 mm. Each element in the array will have ~63 holes that are ~0.5 mm in diameter. Each of the detector array elements will be isolated from the adjacent detector element by a small (~0.13 mm) non-conducting gap. The holes in each detector array element must precisely align when assembled into a linear array (Figure below). An architecture that keeps the array assembled as a unit that is precisely aligned and rigid is necessary. Alternate detectors will be electrically connected to form two inputs to an existing differential pre-amplifier. Design considerations are to maximize transmittance (total area of holes across the surface compared to total area) with a goal of a minimum of 60%, and minimize capacitance of each of the two detector array channels. The feasibility of using closely spaced hexagonal holes to increase the transmittance should be evaluated. All materials must be vacuum compatible (10^{-5} Torr) and the array must mount into an existing linear array charge detection mass spectrometer for performance testing. Integration of a multi-element ion accelerator (TBD) may be required to replace existing accelerator for installation. Construction methods, including CNC, EDM and 3D printing should be considered.

